

## A0 (PyTorch layer)

## A1 (framework layer)

grad-norm  
clip-grad  
no-nan-inf  
ckpt-rng  
step-mono  
init-lr  
replica-eq  
layer-cnt  
ep-group  
router-tok  
jitter-dt  
residual

|                 |           |   |       |   |   |   |   |        |   |   |   |   |        |                    |  |     |  |
|-----------------|-----------|---|-------|---|---|---|---|--------|---|---|---|---|--------|--------------------|--|-----|--|
| onboarding<br>↓ | Megatron  | ★ | ★     | ★ | ★ |   |   | ★      | ★ | ★ | ★ | ★ |        | 150 LoC<br>★9.3 X0 |  |     |  |
|                 | DeepSpeed |   |       |   |   |   |   |        | ✗ |   |   |   |        | 50 LoC<br>★0.7 X1  |  |     |  |
|                 | OLMo      |   |       |   |   |   | ★ |        |   |   |   |   | ★      | 150 LoC<br>★2.7 X0 |  |     |  |
|                 | OLMo-core |   |       |   |   | ★ |   | ✗      |   |   |   |   |        | 150 LoC<br>★1.9 X1 |  |     |  |
|                 | FSDP      |   |       |   |   |   |   |        |   |   |   |   |        | 30 LoC<br>★0.6 X0  |  |     |  |
|                 |           | ★ | mined |   |   |   |   | reused |   |   |   | ✗ | failed |                    |  | N/A |  |